

UK PATENT APPLICATION

SYSTEM AND METHOD FOR AUTONOMOUS CROSS-DOMAIN CAUSAL DISCOVERY AND BUSINESS GENERATION USING ENSEMBLE PREDICTION AND TEST-DRIVEN DEVELOPMENT

ABSTRACT

A computer-implemented system for autonomous cross-domain causal discovery and business generation comprising: a pluggable data ingestion pipeline collecting behavioural and macroeconomic time-series data from heterogeneous sources via a data source registry; a correlation engine computing pairwise associations using multiple correlation coefficients; a bidirectional Granger causality module determining temporal causal direction via frequency-adaptive lag selection; an extensible multi-model ensemble prediction engine combining Granger causality, ordinary least squares regression, and autoregressive integrated moving average forecasts with per-model accuracy tracking and automatic weight calibration from historical accuracy; a cross-layer enforcement architecture constraining predictions from behavioural signal variables to macroeconomic target variables; an exploitation scoring module ranking opportunities; an autonomous test-driven development pipeline converting statistical signals into deployed web applications, with AI-powered specification generation consulting commercial intelligence context and a validation phase producing verification artefacts confirming syntactic correctness and functional compliance; a deployment module deploying generated applications without human intervention; a commercial intelligence module aggregating deployment revenue and engagement metrics to score configuration confidence and enrich future build specifications; and a build-iterate loop diagnosing failures and spawning corrective builds.